Recitation 4

Math 311 Fall 2026
Rutgers University

Exercise 1: Let $x_1 > 0$ and define

$$x_{n+1} = \frac{1}{2}(x_n + \frac{3}{x_n}) \qquad \forall n \in \mathbb{N}.$$

(a) Since $x_1 > 0$, suppose $x_n > 0$ and so

$$x_{n+1} = \frac{1}{2}(x_n + \frac{3}{x_n}) > 0.$$

Thus, $x_n > 0$ for all $n \in \mathbb{N}$ by induction. Then

$$\begin{aligned} x_{n+1} - \sqrt{3} &= \frac{1}{2}\left(x_n + \frac{3}{x_n}\right) - \sqrt{3} \\ &= \frac{x_n^2 + 3 - 2\sqrt{3}x_n}{2x_n} \\ &= \frac{\left(x_n - \sqrt{3}\right)^2}{2x_n}. \end{aligned}$$

Since the numerator is squared, it is non-negative, and we already showed that the $x_n > 0$, so $2x_n > 0$ as well. Thus $x_{n+1} - \sqrt{3} \geq 0$, and so

$$x_n \geq \sqrt{3}.$$

for every $n \geq 2$.

(b) Consider

$$\begin{aligned} x_{n+1} - x_n &= \frac{1}{2}\left(x_n + \frac{3}{x_n}\right) - x_n \\ &= \frac{3}{2x_n} - \frac{1}{2}x_n \\ &= \frac{3 - x_n^2}{2x_n}. \end{aligned}$$

Since $x_n \geq \sqrt{3}$ for every $n \geq 2$, we have

$$3 - x_n^2 \leq 0 \quad \text{and} \quad 2x_n > 0.$$

Then

$$x_{n+1} \leq x_n.$$

Hence the sequence is monotonically decreasing and bounded below, so by the Monotone Convergence Theorem, $\{x_n\}$ converges to some limit L. Taking the limits of the recurrence we have

$$L = \frac{1}{2}\left(L + \frac{3}{L}\right).$$

Which simplifies to

$$L^2 = 3.$$

Therefore,

$$\{x_n\} \to \sqrt{3}.$$